

Effective stability at the symmetric projection-cone endpoint

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Abstract

Let $a(K)$ be the affine projection-cone invariant, and let $\mathcal{E}_d$ be the class of linear images of Cartesian products of centrally symmetric convex bodies of dimensions one and two. For every centrally symmetric full-dimensional convex body in $\mathbb{R}^d$, $d \geq 3$, we prove $D_{\text{BM}}(K, \mathcal{E}_d) - 1 \leq d^{15}(\frac{1}{2} - a(K))^{1/(6d)}$. The estimate imposes no regularity or discreteness assumption. Its proof quantitatively excludes long near-half-mass cofactor circuits by a contact-slab estimate, then constructs a matching of line and plane normal blocks and recovers an actual product using mixed volumes. Corner-truncated cubes have upper-end deficit of order t^d and distance at least $t/(5d^2)$ from the entire equality class, excluding every universal exponent greater than $1/d$.

1. The theorem and explicit constants

Let $K \subset \mathbb{R}^d$ be any full-dimensional centrally symmetric convex body, with its center translated to zero. There is no polytope, smoothness, strict-convexity, absolute-continuity, or positive-curvature assumption. Retain the entry005 invariant

$$a(K) = \left(\frac{d}{d+1}\right)^d \frac{R(\mathcal{P}K)}{R(K)} - 1, \quad R(K) = \frac{|\Pi K|}{|K|^{d-1}}.$$

Let $\mathcal{E}_d$ consist of all linear images of Cartesian products of centrally symmetric factors of dimensions one and two. For centered symmetric bodies use

$$d_{\text{BM}}(K, L) = \inf\{\lambda \geq 1 : TL \subseteq K \subseteq \lambda TL, T \in \text{GL}(d)\}, \quad D(K, \mathcal{E}_d) = \inf_{E \in \mathcal{E}_d} d_{\text{BM}}(K, E).$$

Write $\delta = \frac{1}{2} - a(K)$, which is nonnegative by v3.

Theorem A. Explicit power stability for the entire equality class. For $d = 1, 2$, $D(K, \mathcal{E}_d) = 1$. For every $d \geq 3$, define

$$\begin{aligned} N &= d+1, & \kappa_d &= \frac{\pi^{d/2}}{\Gamma(1+d/2)}, & a_d &= d^{-2d}, \\ Z_d &= \frac{2^{d+9} N \sqrt{d} \binom{d}{3}}{\kappa_d}, & F_d &= 13N + 3Z_d, & B_d &= \frac{4NF_d}{a_d}, \\ L_d &= 6d^3 a_d^{-4} B_d^{1/4}, & H_d &= d(\sqrt{d}+1)L_d, & C_d &= 2d(2dH_d)^{1/d}. \end{aligned}$$

Then, for every such K ,

$$\boxed{D(K, \mathcal{E}_d) - 1 \leq C_d \delta^{1/(6d)}} \tag{1.1}$$

In particular, $C_d \leq d^{15}$, so the following simple numerical version holds in every dimension $d \geq 3$:

$$\boxed{D(K, \mathcal{E}_d) - 1 \leq d^{15}(\frac{1}{2} - a(K))^{1/(6d)}} \tag{1.1a}$$

Equivalently, $\delta \leq (\varepsilon/d^{15})^{6d}$ guarantees a Banach–Mazur factor at most $1 + \varepsilon$. All constants are numerical functions of dimension. They are deliberately conservative. A more informative small-deficit bound is

$$D(K, \mathcal{E}_d) \leq \frac{1}{1 - (2dH_d\delta^{1/6})^{1/d}}, \quad (1.2)$$

whenever

$$0 < \delta \leq \delta_d^* := \left(\frac{2^{-d}}{2dH_d} \right)^6. \quad (1.3)$$

At zero deficit the v4 equality classification applies, or the zero-defect versions of the lemmas below give the same conclusion directly.

Theorem B. A dimensional obstruction. For $d \geq 3$ and $0 < t \leq 1$, set

$$K_t = [-1, 1]^d \cap \{x : |x_1 + \cdots + x_d| \leq d - t\}, \quad c_d = \frac{1}{2} - 2^{1-d}.$$

Then

$$1 + \frac{t}{5d^2} \leq D(K_t, \mathcal{E}_d) \leq \frac{d}{d - t}, \quad (1.4)$$

and

$$\frac{1}{2} - a(K_t) = c_d \frac{2^d - 2t^{d-1}/(d-1)!}{d2^d - 2t^d/(d-1)!} \frac{2t^d}{(d-1)!2^d + 2(d-1)t^{d-1}} \sim \frac{c_d}{d!2^{d-1}} t^d. \quad (1.5)$$

Consequently no universal fixed-dimensional power exponent greater than $1/d$ is possible. No positive exponent independent of dimension can work, even when its constants are allowed to depend on dimension. The optimal exponent remains between $1/(6d)$ and $1/d$; neither theorem identifies it.

The proof combines three effective estimates: a cone-specific contact-slab bound, a quantitative matching of basis coordinates, and a mixed-volume containment argument. The hypothesis and equality class differ from those of the lower-end simplex stability problem.

2. Cone law, contact identity, and the exact upper-end defect

Sections 2–5 assume $d \geq 3$; dimensions one and two were settled in the statement.

The v3 cone law is the pushforward of $h_K(u)dS_K(u)/(d|K|)$ by $u \mapsto u/h_K(u)$. Denote it by ν_K . For independent samples define

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|, \quad B = \mathbb{E} \left| \det \begin{pmatrix} X_1 & \cdots & X_N \\ 1 & \cdots & 1 \end{pmatrix} \right|.$$

We import the v3 identity $a(K) = B/(NA)$, affine invariance, and the balanced Rademacher inequality. These hold for arbitrary convex bodies, by the surface-area formulation and the continuity argument proved there. The law is even for symmetric K .

Put K in symmetric John position:

$$B_2^d \subseteq K \subseteq \sqrt{d}B_2^d. \quad (2.1)$$

Then ν_K is supported in the unit ball and $|K| \geq \kappa_d$. The law can also be sampled by a regular boundary contact $Y \in \partial K$, with density

$$\frac{\langle Y, n_K(Y) \rangle}{d|K|} d\mathcal{H}^{d-1}(Y), \quad X = \frac{n_K(Y)}{\langle Y, n_K(Y) \rangle}.$$

The nonregular boundary has surface measure zero. The divergence theorem gives

$$\langle Y, X \rangle = 1, \quad \mathbb{E}[YX^\top] = I/d. \quad (2.2)$$

For every unit vector v , since $|Y| \leq \sqrt{d}$,

$$\mathbb{E}|\langle v, X \rangle| \geq \frac{1}{d\sqrt{d}}.$$

Conditioning on successive samples and using distance to their span, this implies

$$A = \mathbb{E}\|X_1 \wedge \cdots \wedge X_d\| \geq d^{-3d/2} \geq a_d. \quad (2.3)$$

Indeed, for every fixed proper subspace choose a unit vector perpendicular to it; the conditional expected distance of the next sample to that subspace is at least $d^{-3/2}$. Multiplying these lower bounds proves (2.3), with dependent partial tuples contributing zero harmlessly. Hadamard gives $A \leq 1$.

For an N -tuple, let c_i be its signed horizontal cofactors and put $M = \sum |c_i|$. Their relation is $\sum c_i X_i = 0$. Since the points lie on ∂K° , the Minkowski norm with unit ball K° gives $|c_i| \leq M/2$. On $M > 0$, sort the normalized magnitudes as $t_1 \geq \cdots \geq t_N$, and put

$$\mathcal{D}(t) = \frac{1}{2} - \mathbb{E}_\epsilon \left| \sum_i \epsilon_i t_i \right|, \quad g = \frac{1}{2} - t_1.$$

Define expressions involving t to be zero on $M = 0$. Sign averaging the lifted determinant, using evenness, gives the exact identity

$$\Delta := \mathbb{E}[M\mathcal{D}(t)] = NA\delta, \quad \mathbb{E}M = NA \leq N. \quad (2.4)$$

This includes singular tuples. Let $\phi = Mt_4$ be the fourth-largest absolute cofactor, set to zero if all cofactors vanish.

3. Robust balanced equality and the cone-specific half-mass gate

Lemma 3.1. For balanced normalized coefficients as above,

$$\mathcal{D}(t) \geq \min\{t_4/2, g/4\}. \quad (3.1)$$

Their ℓ_1 -distance to the entire coefficient equality set (at most three nonzero entries, or one entry equal to one half) is at most $8(N-3)\mathcal{D}(t)$.

Proof. On the capped simplex $\{s_i \geq 0, \sum s_i = 1, s_i \leq 1/2\}$, the convex function $\mathbb{E}_\epsilon |\sum \epsilon_i s_i|$ is at most one half by the v3 inequality. Let $y = (1/4, 1/4, 1/4, 1/4, 0, \dots)$, whose value is $3/8$. Set $\alpha = \min(4t_4, 2g)$. If positive, $\alpha < 1$, and $(t - \alpha y)/(1 - \alpha)$ remains in the capped simplex: nonnegativity uses $\alpha \leq 4t_4$; for the first four entries the upper cap is equivalent to $t_i + \alpha/4 \leq 1/2$; for the others it follows from $\alpha \leq 2g$. Convexity yields a defect at least $\alpha/8$, proving (3.1). If $\alpha = 0$, (3.1) is immediate. To reach the half-mass branch, increase t_1 by g and subtract that mass from the other entries, costing $2g$. To reach three-point support, remove the tail of mass $q \leq (N-3)t_4$ and redistribute it in the first three entries without exceeding the cap, costing $2q$. Their available capacity is $1/2 + q$. Use the first construction when the minimum in (3.1) is $g/4$, and the second otherwise. This proves the distance bound. $\square$

The half-mass branch cannot be dropped: $(1/2, 1/6, 1/6, 1/6)$ has zero defect and four positive entries. Exact exposedness in v4 removes it at equality, but does not supply a quantitative margin. We instead use contact geometry.

Lemma 3.2. Corner-slab surface area. If $Q \subseteq [-1, 1]^d$ is any convex body, the regular boundary part where some three coordinates are within $r > 0$ of independently chosen signed cube faces has area at most

$$d2^{d+1} \binom{d}{3} r^2. \quad (3.2)$$

Proof. Fix a triple and its signs and write $\ell = \min(r, 2)$. Each coordinate projection of the relevant boundary part is contained in a box of $(d-1)$ -volume at most $2^{d-3}\ell^2$. If the deleted coordinate is outside the triple, the sharper bound is $2^{d-4}\ell^3$, which is no larger. For any measurable boundary part H , the area formula and convexity give

$$\int_H |n_i| d\mathcal{H}^{d-1} \leq 2|\pi_i H|.$$

Almost every fiber has at most an upper and a lower boundary point with nonzero normal component. Parallel boundary segments have zero such component. Since $\sum_i |n_i| \geq 1$, summing gives area at most $d2^{d-2}r^2$ for one signed triple. There are $8\binom{d}{3}$ choices, proving (3.2), also when $r \geq 2$. $\square$

Lemma 3.3. Weighted exclusion of long near-half-mass circuits. For $0 < \eta \leq 1/4$ and $\rho > 0$,

$$\mathbb{E}[M \mathbf{1}_{\{t_4 > \rho, g \leq \eta\}}] \leq Z_d(\eta/\rho)^2. \quad (3.3)$$

The event is understood on $M > 0$, as already implied by $t_4 > \rho$.

Proof. Take a union over the distinguished index k with $|c_k|/M \geq 1/2 - \eta$, and condition on the other d samples, arranged as columns of a matrix U . If $\det U = 0$, this event at k is impossible. Otherwise put $D_0 = |\det U|$, $X_k = U\alpha$, and $S = \sum |\alpha_j|$. Cofactor identities give

$$|c_k| = D_0, \quad |c_j| = D_0|\alpha_j|, \quad M = D_0(1 + S), \quad \frac{|c_k|}{M} = \frac{1}{1 + S}.$$

On this event $M \leq 4D_0$ and $S - 1 \leq 8\eta$. The condition $t_4 > \rho$ supplies at least three other coordinates with $|\alpha_j| > \rho$.

Set $Q = U^\top K$. Because the columns of U belong to K° , $Q \subseteq [-1, 1]^d$ and $|Q| = D_0|K|$. Under a primal linear map T , normalized cone mass is preserved and the scaled normal becomes $T^{-\top}X$; hence $\nu_Q = (U^{-1})_*\nu_K$. This follows directly from the regular-boundary formulas

$$n' = \frac{T^{-\top}n}{|T^{-\top}n|}, \quad dH' = |\det T| |T^{-\top}n| dH, \quad h' = h/|T^{-\top}n|.$$

For every regular contact $z \in \partial Q$ generating the conditional sample α , $\langle \alpha, z \rangle = 1$, so

$$\sum_j |\alpha_j| (1 - \operatorname{sgn}(\alpha_j) z_j) = S - 1 \leq 8\eta.$$

The three substantial coordinates therefore put z within $r = 8\eta/\rho$ of three signed cube faces. Lemma 3.2 bounds their surface area. The boundary cone density on Q is $\langle z, n_Q(z) \rangle / (d|Q|) \leq \sqrt{d}/(dD_0|K|)$, so the conditional probability is at most

$$\frac{2^{d+1}\sqrt{d}\binom{d}{3}}{D_0|K|} (8\eta/\rho)^2.$$

Multiplication by $M \leq 4D_0$ cancels the determinant exactly. Integrate the conditioning variables, sum over N indices, and use $|K| \geq \kappa_d$. The resulting constant is Z_d . $\square$

This cancellation treats arbitrarily ill-conditioned conditional matrices. The estimate uses regular primal contacts, not strong exposedness of polar points, so atoms caused by flat faces and nonatomic cone laws are both covered.

Lemma 3.4. Effective short-cofactor estimate.

$$\boxed{\mathbb{E}\phi \leq F_d \delta^{2/3}}. \quad (3.4)$$

Proof. First suppose $0 < \delta < 1/4$. For $\rho \in [\delta, 1/4]$, put $\eta = \delta^{1/3} \rho^{2/3} \leq \rho$. In the weighted tail $T(\rho) = \mathbb{E}[M \mathbf{1}_{\{t_4 > \rho\}}]$, the part with $g > \eta$ costs at most $4\Delta/\eta$ by Lemma 3.1. The other part costs at most $Z_d(\eta/\rho)^2$ by Lemma 3.3. Using (2.4),

$$T(\rho) \leq (4N + Z_d) \delta^{2/3} \rho^{-2/3}.$$

Since $t_4 \leq 1/4$, Tonelli gives $\mathbb{E}\phi = \int_0^{1/4} T(\rho) d\rho$. The interval below δ costs at most $N\delta \leq N\delta^{2/3}$; the remaining integral of $\rho^{-2/3}$ is at most 3. This gives (3.4). If $\delta \geq 1/4$, use $\mathbb{E}\phi \leq N/4 \leq N\delta^{2/3}$. If $\delta = 0$, for fixed $\rho > 0$ let $\eta \downarrow 0$ in the same split, with $\eta \leq \min(1/4, 2\rho)$, to get $T(\rho) = 0$. Tonelli proves (3.4) at zero without compactness. $\square$

4. A quantitative matching of line and plane blocks

Lemma 4.1. Let ν be any Borel probability law supported in the unit ball, with $A \geq a$, $0 < a \leq 1$, and suppose its expected fourth absolute cofactor is at most ε . Put $q = 4(d+1)\varepsilon/a$. If $0 < q \leq 1$, there are a basis matrix U and a partition of its coordinate indices into singletons and pairs, with coordinate block union $\mathcal{W}$, such that

$$\begin{aligned} |\det U| &\geq a/2, \quad \|U\| \leq \sqrt{d}, \quad \|U^{-1}\| \leq 2\sqrt{d}/a, \\ \mathbb{E} \text{dist}(U^{-1}X, \mathcal{W}) &\leq 6d^3 a^{-4} q^{1/4}. \end{aligned} \quad (4.1)$$

For $\varepsilon = 0$, the same conclusion holds with zero distance. All statements include nonatomic laws and singular sampled tuples.

Proof. For a deterministic candidate basis $b = (b_1, \dots, b_d)$, define its section cost

$$Q(b) = \int \phi(b, x) d\nu(x) + \sum_{j=1}^d \iint \phi(b_1, \dots, \widehat{b}_j, \dots, b_d, x, y) d\nu(x) d\nu(y).$$

Tonelli gives $\mathbb{E}_b Q(b) \leq (d+1)\varepsilon$. Since $|\det b| \leq 1$, the event $|\det b| \geq a/2$ has probability at least $a/2$. Markov gives $\mathbb{P}(Q > q) \leq a/4$. Thus choose $U = (b_i)$ with $D := |\det U| \geq a/2$ and $Q(U) \leq q$. The norm bounds follow from unit-length columns and inverse rows, each of length at most $1/D$. Cramer's rule also gives $|\alpha_i| \leq 1/D$ for $\alpha = U^{-1}X$.

Let s_3 be the third-largest coordinate magnitude. The absolute cofactors of $(b_1, \dots, b_d, X)$ are D times $(1, |\alpha_1|, \dots, |\alpha_d|)$. If $s_3 \leq 1$, its fourth largest is at least Ds_3 ; otherwise it is at least D , and $s_3 \leq 1/D$. In both cases

$$\mathbb{E}s_3 \leq q/D^2. \quad (4.2)$$

Fix $0 < t \leq 1$ and set $r = Dt^2/2$. Break coordinate ties by index. For each pair ij , let E_{ij} be the event that its coordinates are the largest two, both have magnitude at least t , and $s_3 \leq r$. Write $p_{ij} = \nu(E_{ij})$.

For distinct i, j, k , take $x \in E_{ij}$, $y \in E_{jk}$, with coordinates α, β . In the tuple consisting of the basis except b_j , followed by x, y , the cofactors attached to x, y, b_i, b_k have, up to signs, the values

$$D\beta_j, \quad D\alpha_j, \quad D(\alpha_i\beta_j - \alpha_j\beta_i), \quad D(\alpha_k\beta_j - \alpha_j\beta_k).$$

These are determinant identities without a rank assumption. The first two have magnitude at least Dt . Each of the last two has magnitude at least $D(t^2 - r/D) = Dt^2/2$, since the unwanted coordinate is at most r and the other factor at most $1/D$. Hence

$$p_{ij}p_{jk} \leq \frac{2q}{Dt^2}.$$

Pairs with mass exceeding $w = \sqrt{2q/D}/t$ form a matching. Take its pairs as plane blocks and every remaining index as a line block. Discarded pairs have total mass at most $\binom{d}{2}w$.

Outside the discarded-pair events and $\{s_3 > r\}$, a point is within $\sqrt{d}t$ of its block: if its second coordinate is below t , use the block containing its largest coordinate; otherwise use its matched pair, whose remaining coordinates are at most $r \leq t$. Every point has coordinate norm at most $\sqrt{d}/D$. By (4.2) and Markov,

$$\mathbb{E} \text{dist}(\alpha, \mathcal{W}) \leq \sqrt{d}t + \frac{\sqrt{d}}{D} \left[\binom{d}{2} \frac{\sqrt{2q/D}}{t} + \frac{2q}{D^3t^2} \right]. \quad (4.3)$$

Take $t = q^{1/4}$. Since $q \leq 1$ and $D \geq a/2$, the right side is at most

$$\sqrt{d} [1 + 4 \binom{d}{2} a^{-3/2} + 32a^{-4}] q^{1/4} \leq 6d^3 a^{-4} q^{1/4}$$

for $d \geq 3$. This proves (4.1).

At $\varepsilon = 0$, choose $Q(U) = 0$ on the determinant-good event. Equation (4.2) makes every sample two-sparse almost surely. If two positive-mass two-coordinate classes shared an index, the displayed omitted-basis tuple would have four nonzero cofactors on their product, contradicting its zero section cost. Thus the positive-mass pairs form a matching and the coordinate distance is zero. This also proves the zero case. $\square$

The basis is selected with simultaneous control of all required sections. Its condition number is bounded explicitly. No limiting choice of a basis or unique product decomposition is assumed.

5. From approximate normals to an actual product body

Lemma 5.1. Suppose Q is a full-dimensional centered symmetric convex body, $Q \subseteq RB_2^d$, and $\mathbb{R}^d = \bigoplus_j W_j$ is an orthogonal sum. Set $\mathcal{W} = \bigcup_j W_j$ and $P = \prod_j \text{proj}_{W_j} Q$, with m blocks. If $s = \mathbb{E}_{\nu_Q} \text{dist}(X, \mathcal{W})$, then

$$|P|/|Q| \leq (1 + R(\sqrt{m} + 1)s)^d. \quad (5.1)$$

Putting $v = R(\sqrt{m} + 1)s$, if $2dv < 1$,

$$(1 - (2dv)^{1/d})P \subseteq Q \subseteq P. \quad (5.2)$$

Proof. The projection product contains Q , is symmetric, and lies in $\sqrt{m}RB_2^d$. Its support function agrees with h_Q on every block. For $x \in \partial Q^\circ$, choose a closest block projection $z \in \mathcal{W}$. Support functions are globally Lipschitz with their circumradii, so

$$0 \leq h_P(x) - 1 = h_P(x) - h_Q(x) \leq R(\sqrt{m} + 1)|x - z|.$$

The cone-law definition, homogeneity, and the first mixed-volume formula give

$$\frac{V(Q[d-1], P)}{|Q|} - 1 = \mathbb{E}_{\nu_Q}[h_P(X) - 1] \leq v.$$

Minkowski's first inequality gives (5.1). Consequently $\theta = 1 - |Q|/|P| \leq 1 - (1+v)^{-d} \leq dv$.

For completeness, a symmetric nested-body cap converts this relative volume loss to containment. Let λ be maximal with $\lambda P \subseteq Q$; then $0 < \lambda \leq 1$. If $\lambda < 1$, a touching supporting direction u and point $p \in P$ satisfy $h_Q(u) = \lambda h_P(u)$, $\langle u, p \rangle = h_P(u)$. The image

$$\lambda p + (1 - \lambda)(P \cap \{z : \langle u, z \rangle \geq 0\})$$

lies in P by convexity and outside $\text{int } Q$ by support. Symmetry makes its volume $(1 - \lambda)^d |P|/2$. Hence $\theta \geq (1 - \lambda)^d/2$, proving (5.2). No inverse cone-volume uniqueness theorem is used. $\square$

Proof of Theorem A. Use (2.1). Lemma 3.4 gives $\varepsilon = F_d \delta^{2/3}$. By (2.3), apply Lemma 4.1 with $a = a_d$, so $q = B_d \delta^{2/3}$. For $\delta \leq \delta_d^*$, $q \leq 1$: indeed $H_d \geq B_d^{1/4}$, so $B_d(2^{-d}/(2dH_d))^4 \leq 1$. The resulting coordinate law satisfies

$$\mathbb{E} \text{dist}(U^{-1}X, \mathcal{W}) \leq L_d \delta^{1/6}.$$

Apply the primal map $U^\top$. Its body $Q = U^\top K$ lies in dB_2^d , because $\|U\| \leq \sqrt{d}$, and its cone law is the coordinate law just obtained. Its blocks are orthogonal coordinate lines and planes. Lemma 5.1 gives $v \leq H_d \delta^{1/6}$, with a product $P \in \mathcal{E}_d$, proving (1.2). In the range (1.3), the quantity $r = (2dH_d \delta^{1/6})^{1/d}$ is at most $1/2$, so

$$D(K, \mathcal{E}_d) - 1 \leq \frac{r}{1-r} \leq 2(2dH_d)^{1/d} \delta^{1/(6d)}.$$

For larger deficits, a universal elementary bound is $D(K, \mathcal{E}_d) \leq d$. In John position, $d^{-1/2}[-1, 1]^d \subseteq K \subseteq d^{1/2}[-1, 1]^d$, and the cube belongs to $\mathcal{E}_d$. If $\delta \geq \delta_d^*$, the constant in (1.1) satisfies

$$C_d \delta^{1/(6d)} \geq 2d(2dH_d)^{1/d} (\delta_d^*)^{1/(6d)} = d,$$

which covers $D - 1 \leq d - 1$. At zero use the zero cases of the lemmas or v4. For $d = 1, 2$, every symmetric body is itself an allowed factor. This completes the proof. $\square$

For the stated simpler constant, the cube inscribed in the unit ball gives $\kappa_d \geq (2/\sqrt{d})^d$. Using $N \leq 2d$ and $\binom{d}{3} \leq d^3$, one obtains

$$\begin{aligned} Z_d &\leq 2^{10} d^{d/2+9/2}, & F_d &\leq 2^{12} d^{d/2+9/2}, & B_d &\leq 2^{15} d^{(5/2)d+11/2}, \\ H_d &\leq 12 \cdot 2^{15/4} d^{(69/8)d+47/8}, & C_d^d &\leq 2^d (24 \cdot 2^{15/4}) d^{(77/8)d+55/8}. \end{aligned}$$

Since $2^d \leq d^d$ and $24 \cdot 2^{15/4} < 512 \leq d^6$ for $d \geq 3$,

$$C_d^d \leq d^{(85/8)d+103/8} \leq d^{15d},$$

where the last inequality is $35d \geq 103$. This proves (1.1a).

6. Proof of the dimensional obstruction

We give the full-class argument, since comparing only to the cube would not establish a distance lower bound to arbitrary planar-factor products.

Let $D \in \mathcal{E}_d$ and suppose $K_t \subseteq D \subseteq (1+s)K_t$. Any defining Banach–Mazur sandwich can be rescaled into this form, preserving $\mathcal{E}_d$. Write

$$D = F_1 + \cdots + F_m, \quad \mathbb{R}^d = W_1 \oplus \cdots \oplus W_m, \quad F_j \subset W_j, \quad \dim W_j \in \{1, 2\}.$$

The factors are centered symmetric and full-dimensional in their subspaces. Let P_j be the projection onto W_j along the remaining summands, and set $a_{ji} = h_{F_j}(e_i) \geq 0$. Then

$$P_j^2 = P_j, \quad \text{tr } P_j = \dim W_j, \quad \sum_j P_j = I, \quad 1 \leq \sum_j a_{ji} \leq 1 + s. \quad (6.1)$$

Every cube vertex $\sigma \in \{\pm 1\}^d$ except the two all-equal ones belongs to K_t . Since $P_j \sigma \in F_j$, the nonnegative quantities $a_{ji} - \sigma_i(P_j \sigma)_i$ sum to at most s . Therefore

$$|(P_j \sigma)_i - a_{ji} \sigma_i| \leq s. \quad (6.2)$$

For any coordinate k , assign both signs among the other coordinates; this is possible since $d \geq 3$. Both vertices obtained by changing only σ_k are retained. Subtract (6.2) for them to obtain

$$|(P_j - \text{diag}(a_{j1}, \dots, a_{jd}))_{ik}| \leq s. \quad (6.3)$$

This controls all entries even when the original decomposition is badly conditioned.

Suppose $s \leq 1/(4d)$, and write $p_{ji} = (P_j)_{ii}$. Idempotence gives $|p_{ji}(p_{ji} - 1)| \leq (d-1)s^2$. Thus some $z_{ji} \in \{0, 1\}$ satisfies

$$|p_{ji} - z_{ji}| \leq 2(d-1)s^2.$$

Inside $[0, 1]$, this follows from $q(1-q) \geq q/2$ for distance $q \leq 1/2$ to an endpoint; outside the interval it follows directly. The integer trace and sum identities in (6.1), with errors less than one half, give $\sum_i z_{ji} = \dim W_j$ and $\sum_j z_{ji} = 1$. Hence $J_j = \{i : z_{ji} = 1\}$ partitions the indices into sets of size one or two. For $i \notin J_j$, (6.3) gives $a_{ji} \leq s + 2(d-1)s^2$.

Let $u = (1, \dots, 1)$. Choose a retained vertex with plus signs on J_j and a minus sign outside it. Equation (6.2) implies

$$h_{F_j}(u) \geq \sum_i a_{ji} - 3ds - 4d(d-1)s^2 \geq \sum_i a_{ji} - 4ds.$$

Summing gives $h_D(u) \geq d - 4d^2s$, while containment in $(1+s)K_t$ gives $h_D(u) \leq d - t + ds$. Therefore $t \leq 5d^2s$. If $s > 1/(4d)$, the same lower bound follows from $t \leq 1$. This proves the lower part of (1.4) for every product decomposition. The upper part follows from $(1 - t/d)[-1, 1]^d \subseteq K_t \subseteq [-1, 1]^d$.

For the exact deficit, the two removed corner simplices each have volume $t^d/d!$, and every coordinate facet has area $\alpha = 2^{d-1} - t^{d-1}/(d-1)!$. The two new area-normal vectors are $\pm t^{d-1}u/(d-1)!$, with area times support number $\beta = (d-t)t^{d-1}/(d-1)!$. Put

$$V_t = 2^d - 2t^d/d!, \quad D_t = d2^d - 2t^d/(d-1)!, \quad p = \frac{2^d - 2t^{d-1}/(d-1)!}{D_t}, \quad r = \frac{2(d-t)t^{d-1}}{(d-1)!D_t}.$$

The cone law has direction pairs $\pm e_i$, each with total probability p , and $\pm u/(d-t)$, with total probability r . Within pairs signs are uniform; $dp + r = 1$. Direct type enumeration gives

$$A = d!p^{d-1} \left(p + \frac{dr}{d-t} \right).$$

Only N -tuples containing all $d+1$ different direction types have a nonzero Rademacher defect. Tuples with fewer than d types have zero cofactors; those with exactly d types have a repeated parallel direction and a two-term cofactor relation. For the all-distinct tuple, cofactor magnitudes are 1 and d copies of $q = 1/(d-t)$. Averaging the sign multiplying 1 gives $\mathbb{E} \max(1, |q \sum_{i=1}^d \epsilon_i|)$. For $t \leq 1$, only the two all-equal sign patterns exceed 1, so its value is $1 + 2^{1-d}(dq - 1)$. The defect is $c_d t/(d-t)$. The probability of all-distinct types is $N!p^d r$, giving

$$\delta_t = \frac{c_d p r t}{(d-t)p + dr}.$$

Substitute p, r to get (1.5). Also $p \leq 1/d$, hence $\delta_t \leq c_d t^d/(d!2^{d-1})$. Combining this with (1.4) excludes every exponent greater than $1/d$ as $t \downarrow 0$. In dimension three,

$$\delta_t = \frac{t^3(8-t^2)}{8(24-t^3)(4+t^2)}.$$

This proves Theorem B. $\square$

7. Comparison with primary cone-volume and stability literature

The relevant antecedent is Böröczky–Lutwak–Yang–Zhang, [The logarithmic Minkowski problem](#), JAMS 26 (2013), 831–852. Its subspace concentration condition and complementary-subspace clause characterize existence of symmetric cone-volume measures. They do not supply the near-short-circuit estimate (3.4), a quantitative matching into dimensions at most two, or the present invariant’s endpoint.

Böröczky–Henk, [Cone-volume measure and stability](#), Adv. Math. 306 (2017), 24–50, Theorem 1.1 gives complementary-factor stability with containment exponent $1/(5d)$ from almost saturated cone mass on an **exact** proper subspace; its constants and smallness threshold are stated as dimension-dependent, rather than numerical formulas. Its U-functional stability targets parallelotopes. Our equality class includes all symmetric planar factors, and our hypothesis is determinant magnitude, not the U-functional’s independence indicator. Thus that theorem is not an equivalent endpoint result.

The exact-subspace premise cannot be inferred here: for even integers $p \rightarrow \infty$, smooth strictly convex ℓ_p balls converge to the cube and have $a \rightarrow 1/2$, yet their cone-volume measures give zero mass to every proper linear subspace. On each open coordinate octant the Gauss map is a diffeomorphism with positive curvature; the omitted coordinate-zero parts have surface measure zero, proving absolute continuity. Our method instead produces an expected distance to blocks and recovers an actual product by Lemma 5.1.

Böröczky–De, [Stable solution of the logarithmic Minkowski problem in the case of hyperplane symmetries](#), J. Differential Equations 298 (2021), 298–322, obtains quantitative inverse cone-volume estimates under Coxeter symmetry and a tube concentration margin. The reducible-group margin degenerates at concentration on a proper group-invariant product block, and the symmetry hypothesis does not cover arbitrary symmetric planar factors. Their anisotropic-squeeze example also explains

why normalization is necessary. Abdallah–Mérigot, [On the reconstruction of convex sets from random normal measurements](#), gives a surface-area inverse estimate of order $1/d$ with area and rotundity control. Cone-volume data additionally contains the unknown support function; Lemma 5.1 avoids solving that inverse problem.

Böröczky–De also compare a cube with a volume-normalized unconditional body obtained by truncating its vertices: the cone-volume Wasserstein distance is $O_d(\varepsilon)$, whereas the containment loss is $\Omega_d(\varepsilon^{1/d})$. This is a related cap-volume obstruction. Theorem B instead computes the upper-end scalar deficit $1/2 - a(K_t)$ and proves a distance lower bound against every affine product of symmetric one- and two-dimensional bodies. A comparison with one fixed cube would not establish that lower bound.

Saroglou’s [Volumes of projection bodies of some classes of convex bodies](#) is a primary projection-body comparison. Only its publication record and abstract were available; a full-text comparison remains incomplete. The recent Liu–Xiong–Yang mixed-volume stability paper, DOI [10.1016/j.aim.2026.110886](#), is relevant to broader mixed-volume stability questions. No theorem from either of these two papers is imported as a dependency. The present literature comparison is confined to the primary statements described above.

8. Dependencies, reproducibility, and open questions

The affine calculus and cone-law representation are inherited from entry 005 versions 2–3; the exact equality class is the version 4 theorem. The pinned sources are listed below. The quantitative contact-slab, matching, and product recovery arguments in Sections 3–5 are supplied in full here. In particular, the lower-end simplex modulus and the balanced-recursion spectral theorem are separate results and are not inputs to this proof.

The accompanying exact programs test 4,930 balanced rational coefficient multisets, 750 omitted-basis cofactor identities, and 16 corner-truncation facet-minor calculations. They use only integers and rational arithmetic, and explicit exceptions keep the checks active under Python optimization. These finite regressions supplement the written analytic proof. The package records source hashes and separates regenerated outputs from shipped reference reports.

The constants are deliberately coarse. The optimal universal exponent remains between $1/(6d)$ and $1/d$ for $d \geq 3$. Sharpening the matching estimate and the conversion of normal concentration to product containment may reduce the gap.

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- [Sch] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Cambridge University Press, 2014. Support functions, surface area, John normalization, and mixed volumes.

Versions 2–4 credit the projection-volume product and simplex computations inherited from OpenAI family 088, pinned at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. This supplement uses the inherited identities through the explicitly cited entry 005 sources.